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Sem : V

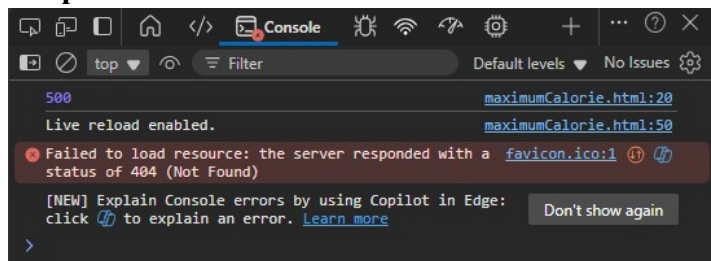
PRN : 23070521225

Part-01

Code 1:

```
1 <!-- Function Declarations
2 Write a function maxCalories(breakfast, lunch) that takes two calorie counts and returns
3 the higher one.
4 This is used to check which meal had more calories.
5 Input: maxCalories(350, 500)
6 Output: 500 -->
7
8 <!DOCTYPE html>
9 <head>
10   <title>Calculate Higher Calorie</title>
11 </head>
12 <body>
13   <script>
14     function maxCalories(breakfast, lunch)
15     {
16       return Math.max(breakfast, lunch);
17     }
18
19     let result = maxCalories(350, 500);
20     console.log(result);
21   </script>
22 </body>
23 </html>
```

Output:



Code 2:

```
1 <!-- Create a function expression stepsFactorial that calculates the factorial of steps (to
2 generate a unique challenge ID for fitness competitions).
3 Input: stepsFactorial(4)
4 Output: 24 -->
5
6 <!DOCTYPE html>
7 <head>
8   <title>Compute Factorial</title>
9 </head>
10 <body>
11   <script>
12     const stepsFactorial = function(steps)
13     {
14       let fact = 1;
15
16       for(let x=1; x<=steps; x++)
17         fact *= x;
18
19       return fact;
20     }
21
22     let result = stepsFactorial(4);
23     console.log(result);
24   </script>
25 </body>
26 </html>
```

24 [stepFactorial.html:23](#)

Live reload enabled. [stepFactorial.html:53](#)

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Output:

```
1 <!-- Write an arrow function checkWorkoutNote(note) that returns true if the note contains
2 the word "Workout".
3 Input: checkWorkoutNote("Today's Workout was intense!")
4 Output: true -->
5
6 <!DOCTYPE html>
7 <head>
8   <title>Check Work Out Note</title>
9 </head>
10 <body>
11   <script>
12     const checkWorkoutNote = (note) =>
13     {
14       return note.includes("Workout");
15     };
16
17     let result = checkWorkoutNote("Today's Workout was intense!");
18     console.log(result);
19   </script>
20 </body>
21 </html>
```

Output:

true [workNote.html:18](#)

Live reload enabled. [workNote.html:48](#)

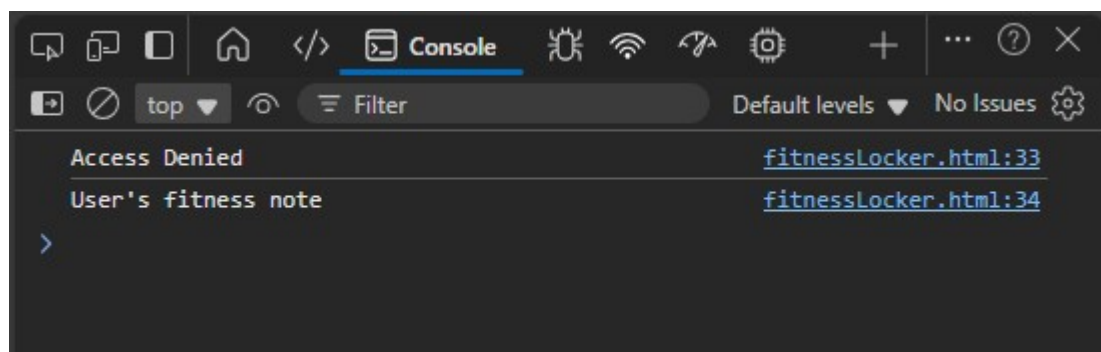
Failed to load resource: the server responded with a status of 404 (Not Found) [favicon.ico:1](#)

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Code 3:

```
fitnessLocker.html > ...
1  <!-- Build a closure fitnessLocker(pin) that stores a user's fitness note.
2  The note should be revealed only if the correct pin is provided.
3  Input:
4
5  const locker = fitnessLocker(&quot;9999&quot;);
6  console.log(locker(&quot;1234&quot;));
7  console.log(locker(&quot;9999&quot;));
8  Output:
9  Access Denied
10 (User's fitness note)
11
12 <!DOCTYPE html>
13 <head>
14   <title>Fitness Locker</title>
15 </head>
16 <body>
17   <script>
18     function fitnessLocker(pin)
19     {
20       const fitnessNote = "User's fitness note";
21
22       return function(enteredPin)
23       {
24         if(enteredPin == pin)
25           return fitnessNote;
26         else
27           return "Access Denied";
28       }
29     }
30
31     const locker = fitnessLocker("9999");
32
33     console.log(locker("1234"));
34     console.log(locker("9999"));
35   </script>
36 </body>
37 </html>
```

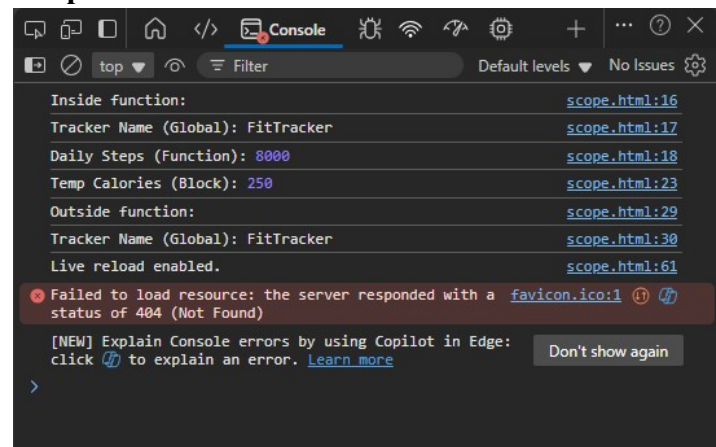
Output:



Code 4:

```
scope.html > ...
1 <!-- Show global, function, and block scope with variables such as trackerName, dailySteps,
2 and tempCalories. -->
3
4 <!DOCTYPE html>
5 <head>
6 |   <title>Scope</title>
7 </head>
8 <body>
9 |   <script>
10 |     let trackerName = "FitTracker"; //Global Scope
11 |
12 |     function stepTracker()
13 |     {
14 |       let dailySteps = 8000; //Function Scope
15 |
16 |       console.log("Inside function:");
17 |       console.log("Tracker Name (Global):", trackerName); //accessible
18 |       console.log("Daily Steps (Function):", dailySteps); //accessible
19 |
20 |       if(dailySteps>5000)
21 |       {
22 |         let tempCalories = 250; //Block Scope
23 |         console.log("Temp Calories (Block):", tempCalories); //accessible inside block
24 |       }
25 |       //console.log(tempCalories); //Error (cannot be accessed outside the block)
26 |     }
27 |
28 |     stepTracker();
29 |     console.log("Outside function:");
30 |     console.log("Tracker Name (Global):", trackerName); //accessible outside the function
31 |     //console.log(dailySteps); //Error (cannot be accessed outside the function)
32 |   </script>
33 </body>
34 </html>
```

Output:



Inside function: [scope.html:16](#)

Tracker Name (Global): FitTracker [scope.html:17](#)

Daily Steps (Function): 8000 [scope.html:18](#)

Temp Calories (Block): 250 [scope.html:23](#)

Outside function: [scope.html:29](#)

Tracker Name (Global): FitTracker [scope.html:30](#)

Live reload enabled. [scope.html:61](#)

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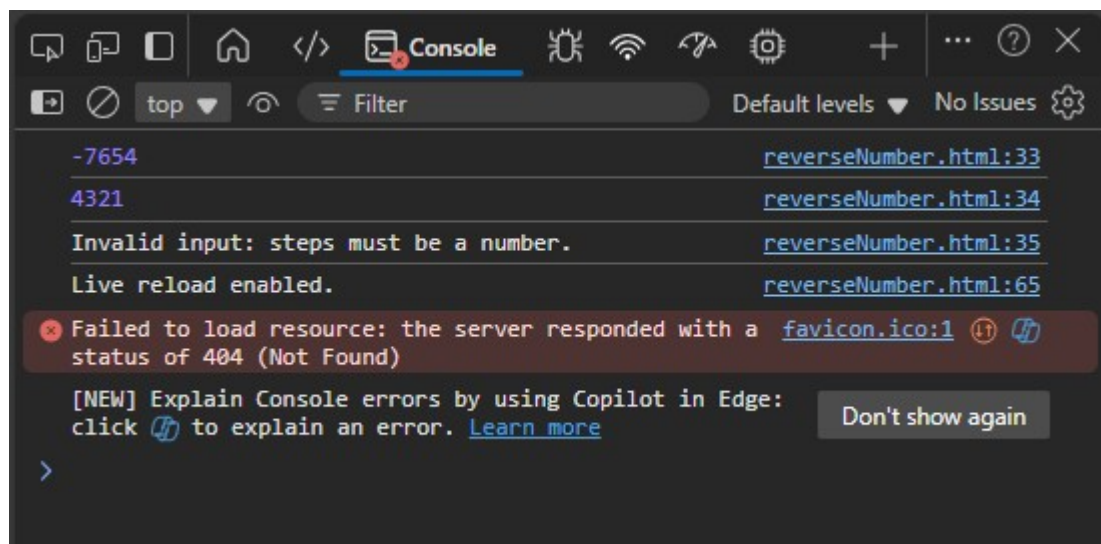
Part-02

Code 1:

```
reverseNumber.html > ...
1 <!-- Write reverseSteps(steps) that works for both positive and negative step counts (though
2 negative is unusual, it's allowed for testing).
3 Use try...catch to handle invalid inputs.
4 Input: reverseSteps(-4567)
5 Output: -7654 -->
6
7 <!DOCTYPE html>
8 <head>
9 | <title>Display Reverse Number</title>
10 </head>
11 <body>
12 | <script>
13 |   function reverseSteps(steps)
14 |   {
15 |     try
16 |     {
17 |       if(typeof steps !== "number" || isNaN(steps))
18 |         throw new Error("Invalid input: steps must be a number.");
19 |
20 |       const sign = Math.sign(steps);
21 |
22 |       let reversed = Math.abs(steps).toString().split("").reverse().join("");
23 |       reversed = parseInt(reversed, 10);
24 |
25 |       return sign * reversed;
26 |     }
27 |     catch(err)
28 |     {
29 |       return err.message;
30 |     }
31 |   }
32 |
33 |   console.log(reverseSteps(-4567));
34 |   console.log(reverseSteps(1234));
35 |   console.log(reverseSteps("abc"));
36 | </script>
37 </body>
38 </html>
```

Output:

```
palindrome.html > ...
1 <!-- Create a function isFitnessPalindrome(id) to check if a fitness ID is a palindrome.
2 Input: isFitnessPalindrome(1221)
3 Output: true -->
4
5 <!DOCTYPE html>
6 <head>
7 | <title>Check Pallindrome</title>
8 </head>
9 <body>
10 | <script>
11 |   function isFitnessPalindrome(id)
12 |   {
13 |     const strId = id.toString();
14 |     const reversed = strId.split("").reverse().join("");
15 |
16 |     return strId == reversed;
17 |   }
18 |
19 |   console.log(isFitnessPalindrome(1221));
20 |   console.log(isFitnessPalindrome(1234));
21 | </script>
22 </body>
23 </html>
```



Output:

Code 2:

```

true                                                                 pallindrome.html:19
false                                                                pallindrome.html:20
Live reload enabled.                                                 pallindrome.html:50
Failed to load resource: the server responded with a status of 404 (Not Found)
[NEW] Explain Console errors by using Copilot in Edge:
click  to explain an error. Learn more
Don't show again
>

workOutPallindrome.html > ...
1  <!-- Extend functionality: accept a workout note string and check if it is a palindrome
2  (ignore spaces and case).
3  Input: &quot;Noon&quot;;
4  Output: true
5
6  <!DOCTYPE html>
7  <head>
8    <title>Work Out Pallindrome</title>
9  </head>
10 <body>
11   <script>
12     function isWorkoutPallindrome(note)
13     {
14       if(typeof note != "string")
15         throw new Error("Note must be a string");
16       const cleaned = note.replace(/\s+/g, "").toLowerCase();
17       const reversed = cleaned.split("").reverse().join("");
18
19       return cleaned == reversed;
20     }
21
22     console.log(isWorkoutPallindrome("Noon"));
23     console.log(isWorkoutPallindrome("Was it a car"));
24     console.log(isWorkoutPallindrome("Race car"));
25   </script>
26 </body>
27 </html>
```

Output:

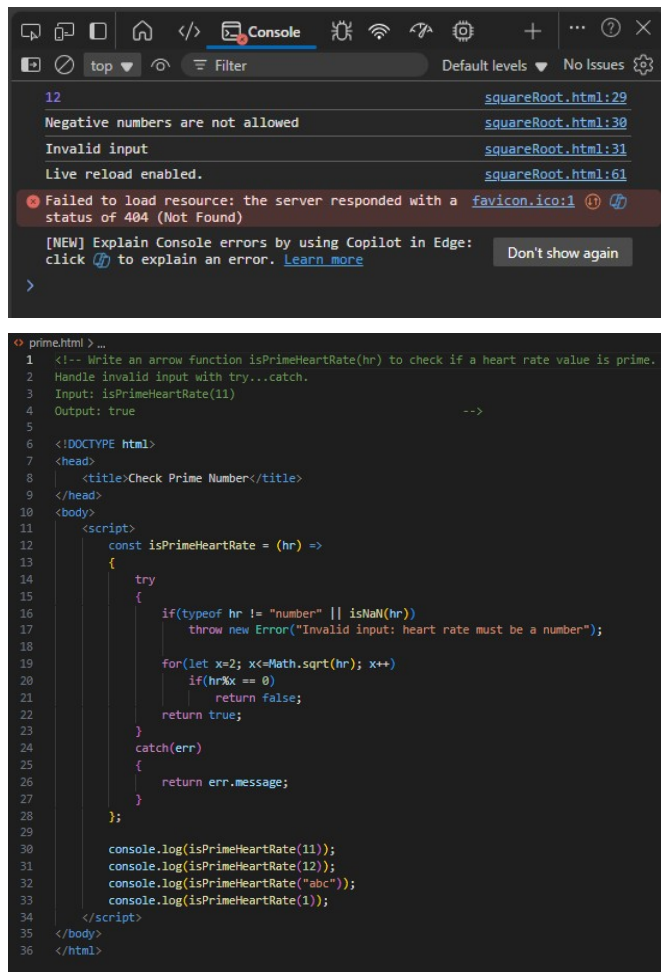
```

true                                                                 workOutPallindrome.html:22
false                                                                workOutPallindrome.html:23
true                                                                 workOutPallindrome.html:24
Live reload enabled.                                                 workOutPallindrome.html:54
Failed to load resource: the server responded with a status of 404 (Not Found)
[NEW] Explain Console errors by using Copilot in Edge:
click  to explain an error. Learn more
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>

squareRoot.html > ...
1  <!-- Implement a function calorieSqrt(calories) that returns the square root of calories burned.
2  If the input is invalid or negative, throw an error and catch it.
3  Input: calorieSqrt(-100)
4  Output: &quot;Error: Negative numbers not allowed&quot;;
5
6  <!DOCTYPE html>
7  <head>
8    <title>Compute Square Root</title>
9  </head>
10 <body>
11   <script>
12     function calorieSqrt(calories)
13     {
14       try
15       {
16         if(typeof calories != "number" || isNaN(calories))
17           throw new Error("Invalid input");
18         if(calories<0)
19           throw new Error("Negative numbers are not allowed");
20
21         return Math.sqrt(calories);
22       }
23       catch(err)
24       {
25         return err.message;
26       }
27     }
28
29     console.log(calorieSqrt(144));
30     console.log(calorieSqrt(-100));
31     console.log(calorieSqrt("abc"));
32   </script>
33 </body>
34 </html>
```

Output:

Code 3:



The screenshot shows the VS Code interface. The top panel displays the 'Console' window with the following messages:

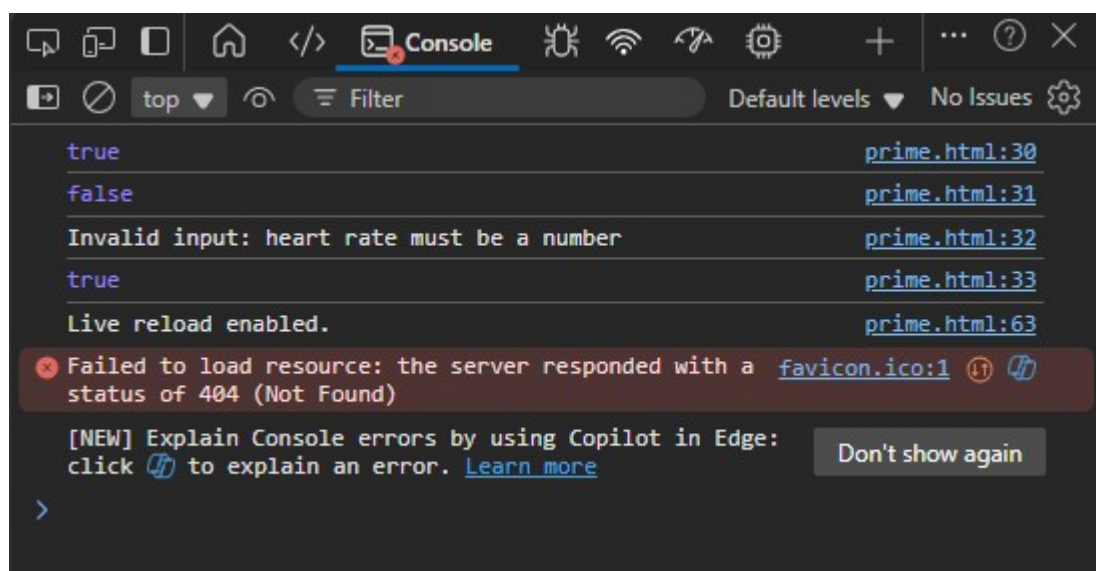
- 12 [squareRoot.html:29](#)
- Negative numbers are not allowed [squareRoot.html:30](#)
- Invalid input [squareRoot.html:31](#)
- Live reload enabled. [squareRoot.html:61](#)
- Failed to load resource: the server responded with a status of 404 (Not Found) [favicon.ico:1](#)

Below the console, a message states: [NEW] Explain Console errors by using Copilot in Edge: click to explain an error. [Learn more](#) Don't show again

The bottom panel shows the code editor with the following code:

```
1 <!-- Write an arrow function isPrimeHeartRate(hr) to check if a heart rate value is prime.
2 Handle invalid input with try...catch.
3 Input: isPrimeHeartRate(11)
4 Output: true -->
5
6 <!DOCTYPE html>
7 <head>
8   <title>Check Prime Number</title>
9 </head>
10 <body>
11   <script>
12     const isPrimeHeartRate = (hr) =>
13     {
14       try
15       {
16         if(typeof hr !== "number" || isNaN(hr))
17           throw new Error("Invalid input: heart rate must be a number");
18
19         for(let x=2; x<=Math.sqrt(hr); x++)
20           if(hr%x == 0)
21             return false;
22         return true;
23       }
24       catch(err)
25       {
26         return err.message;
27       }
28     };
29
30     console.log(isPrimeHeartRate(11));
31     console.log(isPrimeHeartRate(12));
32     console.log(isPrimeHeartRate("abc"));
33     console.log(isPrimeHeartRate(1));
34   </script>
35 </body>
36 </html>
```

Output:



The screenshot shows the VS Code interface with the 'Console' window displaying the output of the code:

- true [prime.html:30](#)
- false [prime.html:31](#)
- Invalid input: heart rate must be a number [prime.html:32](#)
- true [prime.html:33](#)
- Live reload enabled. [prime.html:63](#)
- Failed to load resource: the server responded with a status of 404 (Not Found) [favicon.ico:1](#)

Below the console, a message states: [NEW] Explain Console errors by using Copilot in Edge: click to explain an error. [Learn more](#) Don't show again

Code 4:

Part-03 (Menu Driven Interface)

Code:

```
<> utilityFunctions.html > ...
1  <!-- Develop a menu-driven console application that offers users the following options:
2  1. Compare calories of breakfast vs lunch.
3  2. Generate factorial-based fitness challenge ID.
4  3. Check if a workout note contains "Workout".
5  4. Store and retrieve a fitness note using a PIN (closure).
6  5. Reverse a step count and check if it's a palindrome.
7  6. Check if a workout note is a palindrome (word-based).
8  7. Find square root of burned calories with error handling.
9  8. Check if a heart rate value is prime.
10 Each option must:
11   • Accept input.
12   • Validate using error handling.
13   • Display proper results.
14
15 <!DOCTYPE html>
16 <head>
17 |   <title>Fitness Utility Console Application</title>
18 </head>
19 <body>
20 |   <script>
21 |     // 1. Compare calories of breakfast vs lunch
22 |     function compareCalories(breakfast, lunch)
23 |     {
24 |       try
25 |       {
26 |         if(isNaN(breakfast) || isNaN(lunch))
27 |         |   throw "Calories must be in numbers.";
28 |         return breakfast > lunch
29 |         ? `Breakfast (${breakfast} cal) had more than Lunch (${lunch} cal).`
30 |         : `Lunch (${lunch} cal) had more than Breakfast (${breakfast} cal).`;
31 |       }
32 |       catch(err)
33 |       {
34 |         return `Error: ${err.message}`;
35 |       }
36 |     }
37
```


Code 5:

```
38 // 2. Factorial-based fitness challenge ID
39 function factorialChallengeID(num)
40 {
41     try
42     {
43         if(isNaN(num) || num < 0)
44             throw "Steps must be a non-negative number.";
45         let fact = 1;
46         for(let i=1; i<=num; i++)
47             fact *= i;
48         return `Challenge ID based on ${num} steps = ${fact}`;
49     }
50     catch(err)
51     {
52         return `Error: ${err.message}`;
53     }
54 }
55
56 // 3. Check if workout note contains "Workout"
57 function containsWorkout(note)
58 {
59     try
60     {
61         if(typeof note !== "string")
62             throw "Note must be text.";
63         return note.includes("Workout") ? "Note contains 'Workout'" : "No 'Workout' found.";
64     }
65     catch(err)
66     {
67         return `Error: ${err.message}`;
68     }
69 }
70
71 // 4. Closure: Secure fitness note locker
72 function fitnessLocker(pin)
73 {
74     const note = "Your private fitness note: Keep pushing!";
75     return function(enteredPin)
76     {
77         return enteredPin === pin ? note : "Access Denied";
78     };
79 }
80
81 // 5. Reverse step count and check palindrome
82 function reverseStepsAndCheckPalindrome(steps)
83 {
84     try
85     {
86         if(isNaN(steps))
87             throw "Steps must be a number.";
88         const sign = Math.sign(steps);
89         let reversed = Math.abs(steps).toString().split("").reverse().join("");
90         reversed = parseInt(reversed, 10) * sign;
91         const isPal = Math.abs(steps).toString() === Math.abs(reversed).toString();
92         return `Reversed: ${reversed}, Palindrome: ${isPal}`;
93     }
94     catch(err)
95     {
96         return `Error: ${err.message}`;
97     }
98 }
99
```

Code 6:

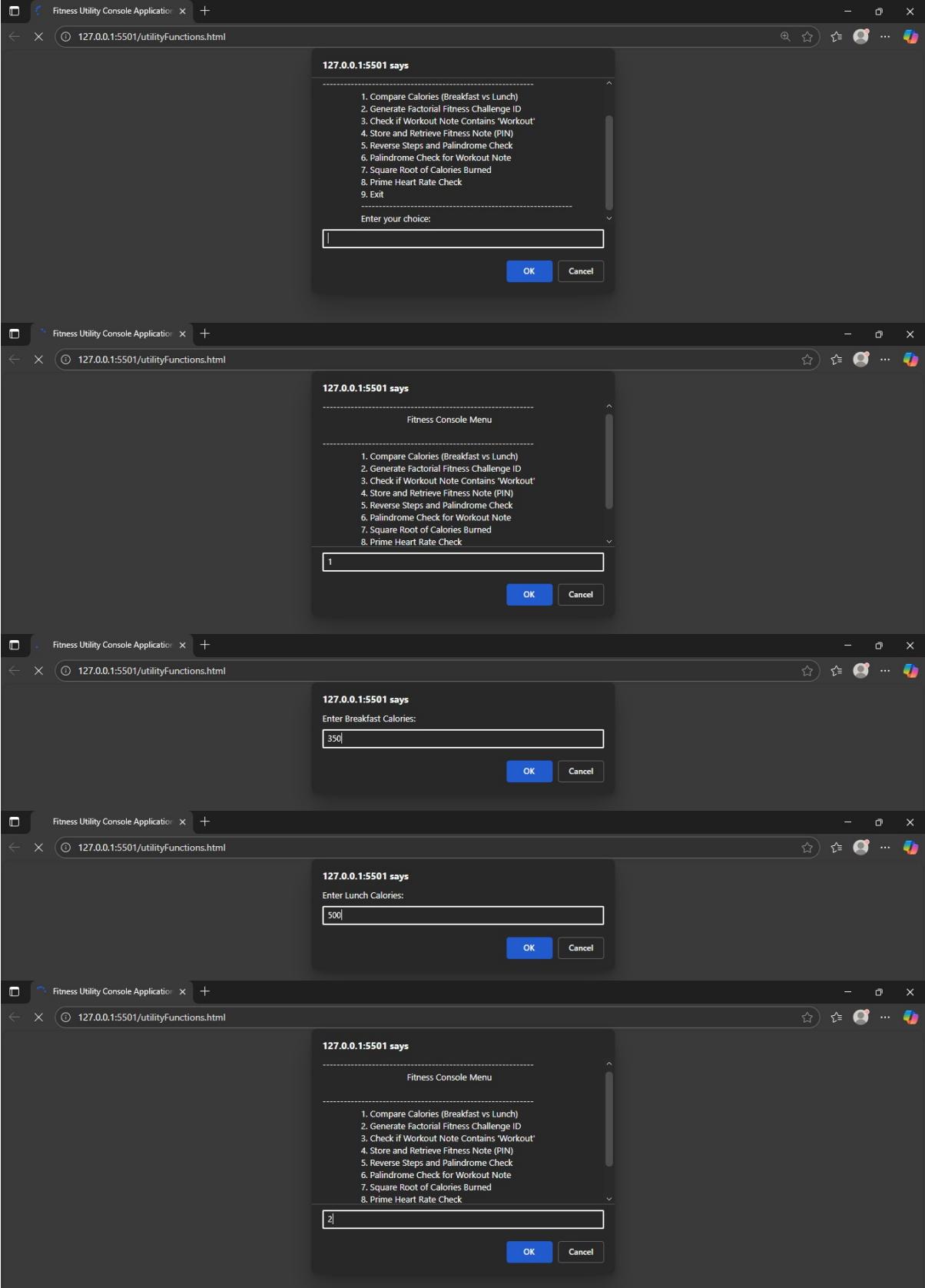
```
100 // 6. Palindrome check for workout note
101 function isNotePalindrome(note)
102 {
103     try
104     {
105         if(typeof note !== "string")
106             throw "Note must be text.";
107         const clean = note.toLowerCase().replace(/^[^a-z0-9]/g, "");
108         const reversed = clean.split("").reverse().join("");
109         return clean === reversed ? "Note is a Palindrome" : "Note is not a Palindrome";
110     }
111     catch(err)
112     {
113         return `Error: ${err.message}`;
114     }
115 }
116
117 // 7. Square root of calories burned
118 function calorieSqrt(calories)
119 {
120     try
121     {
122         if(isNaN(calories))
123             throw "Calories must be a number.";
124         if(calories < 0)
125             throw "Negative calories are not allowed.";
126         return `Square root of ${calories} = ${Math.sqrt(calories)}`;
127     }
128     catch(err)
129     {
130         return `Error: ${err.message}`;
131     }
132 }
133
134 // 8. Prime heart rate check
135 function isPrimeHeartRate(hr)
136 {
137     try
138     {
139         if(isNaN(hr) || typeof hr !== "number")
140             throw "Heart rate must be a number.";
141         for(let i=2; i<=Math.sqrt(hr); i++)
142             if (hr%i === 0)
143                 return `${hr} is NOT prime.`;
144         return `${hr} is PRIME.`;
145     }
146     catch(err)
147     {
148         return `Error: ${err.message}`;
149     }
150 }
151
152 // MENU
153 function menu()
154 {
155     let op;
156     do
157     {
158         op = parseInt(prompt(
159             `
160             |-----|
161             |           | Fitness Console Menu |           |
162             |-----|
163             1. Compare Calories (Breakfast vs Lunch)
164             2. Generate Factorial Fitness Challenge ID
165             3. Check if Workout Note Contains 'Workout'
166             4. Store and Retrieve Fitness Note (PIN)
167             5. Reverse Steps and Palindrome Check
168             6. Palindrome Check for Workout Note
169             7. Square Root of Calories Burned
170             8. Prime Heart Rate Check
171             9. Exit
172             |-----|
173             Enter your choice:
174             `));
175     }
176 }
```

Code 7:

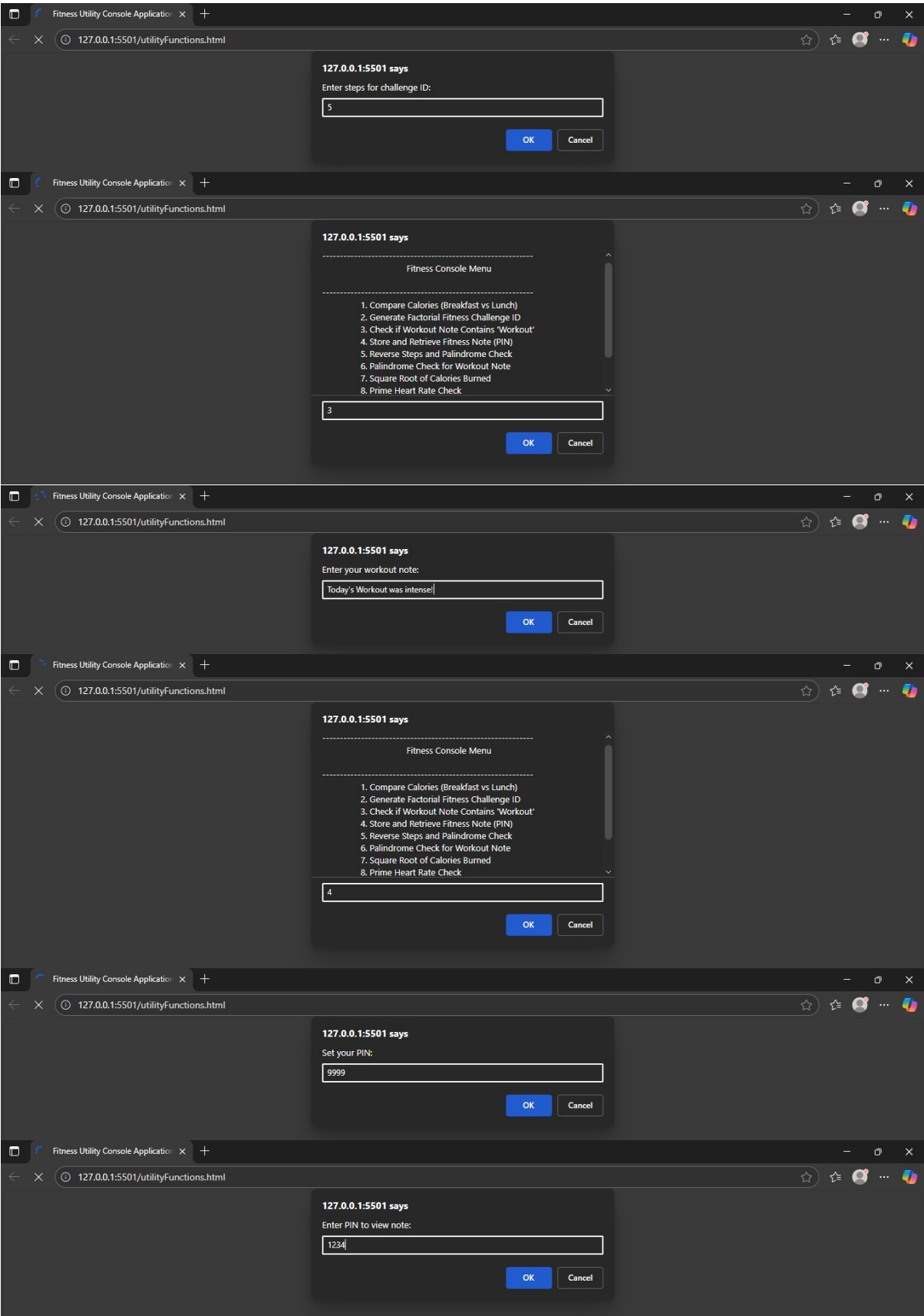
```
175         switch(op)
176         {
177             case 1:
178                 let b = parseInt(prompt("Enter Breakfast Calories:"));
179                 let l = parseInt(prompt("Enter Lunch Calories:"));
180                 console.log(compareCalories(b, l));
181                 break;
182
183             case 2:
184                 let steps = parseInt(prompt("Enter steps for challenge ID:"));
185                 console.log(factorialChallengeID(steps));
186                 break;
187
188             case 3:
189                 let note = prompt("Enter your workout note:");
190                 console.log(containsWorkout(note));
191                 break;
192
193             case 4:
194                 let pin = prompt("Set your PIN:");
195                 let locker = fitnessLocker(pin);
196                 let enteredPin = prompt("Enter PIN to view note:");
197                 console.log(locker(enteredPin));
198                 break;
199
200             case 5:
201                 let sc = parseInt(prompt("Enter step count:"));
202                 console.log(reverseStepsAndCheckPalindrome(sc));
203                 break;
204
205             case 6:
206                 let wn = prompt("Enter a workout note:");
207                 console.log(isNotePalindrome(wn));
208                 break;
209
210             case 7:
211                 let cal = parseInt(prompt("Enter calories burned:"));
212                 console.log(calorieSqrt(cal));
213                 break;
214
215             case 8:
216                 let hr = parseInt(prompt("Enter heart rate:"));
217                 console.log(isPrimeHeartRate(hr));
218                 break;
219
220             case 9:
221                 console.log("Exiting Fitness Console. Stay healthy! ");
222                 break;
223
224             default:
225                 console.log("Invalid choice. Try again.");
226         }
227     }while(op != 9);
228 }
229 menu();
230 </script>
231 </body>
232 </html>
```

Code 8:

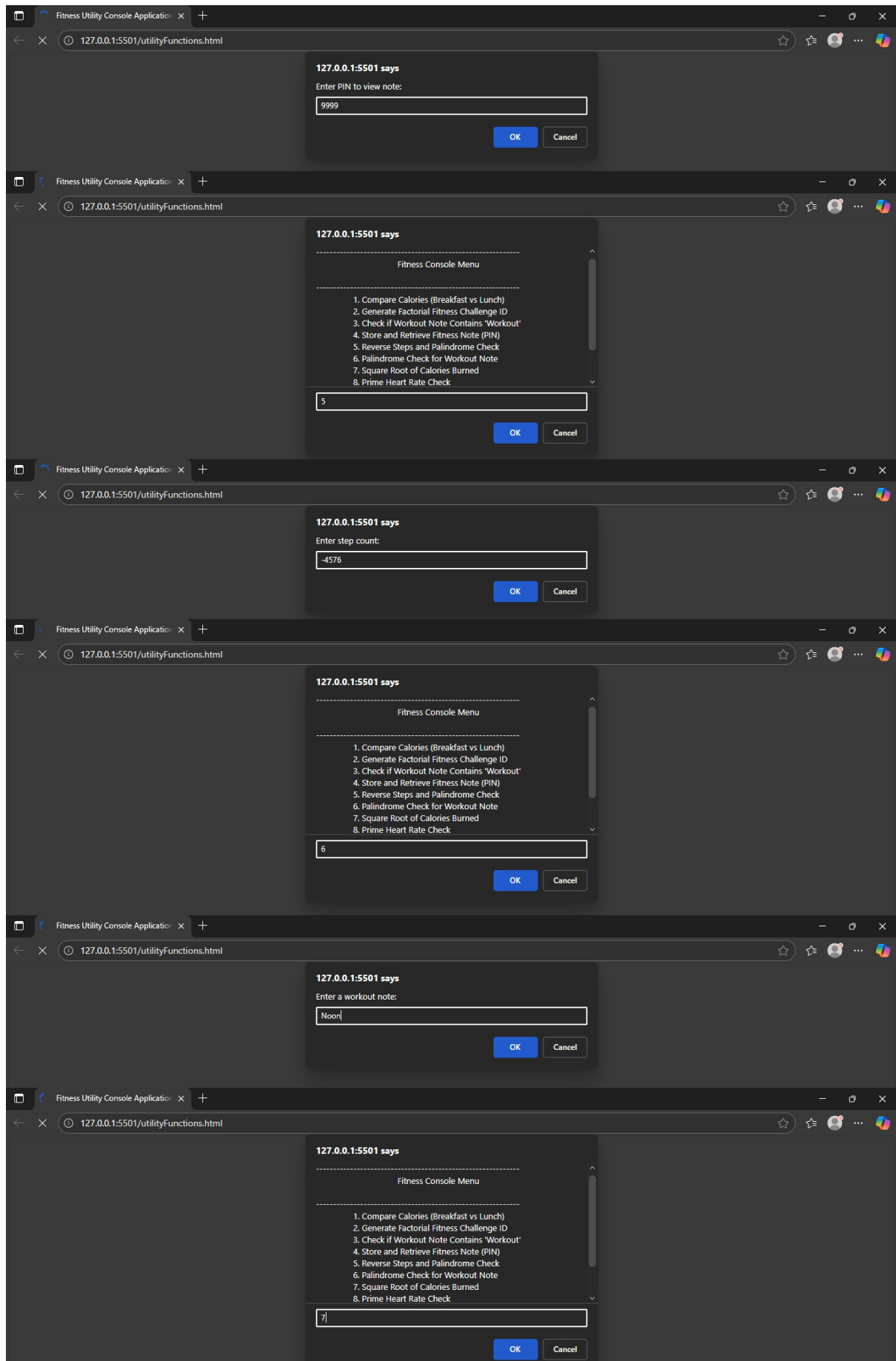
Output:



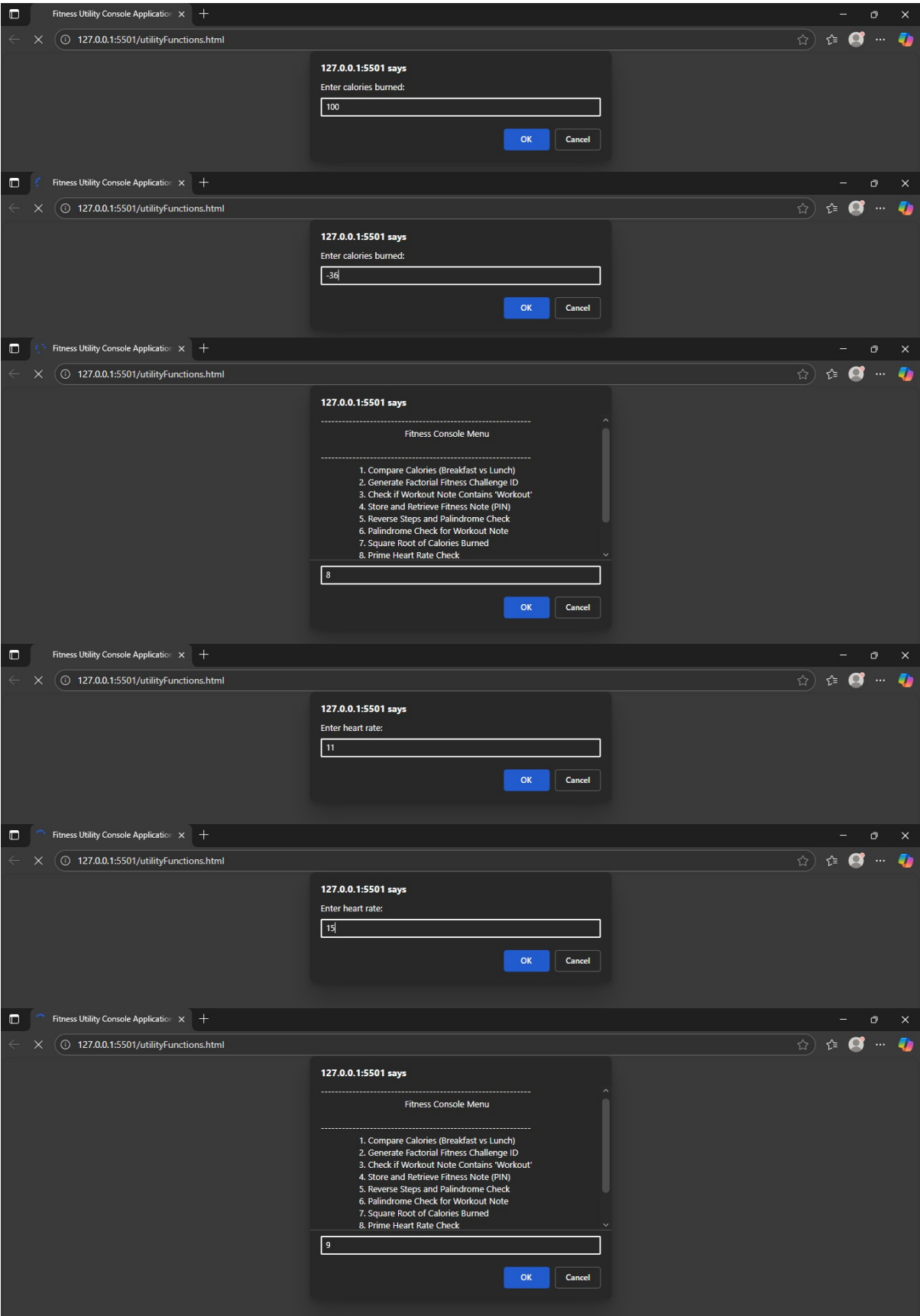
Code 9:



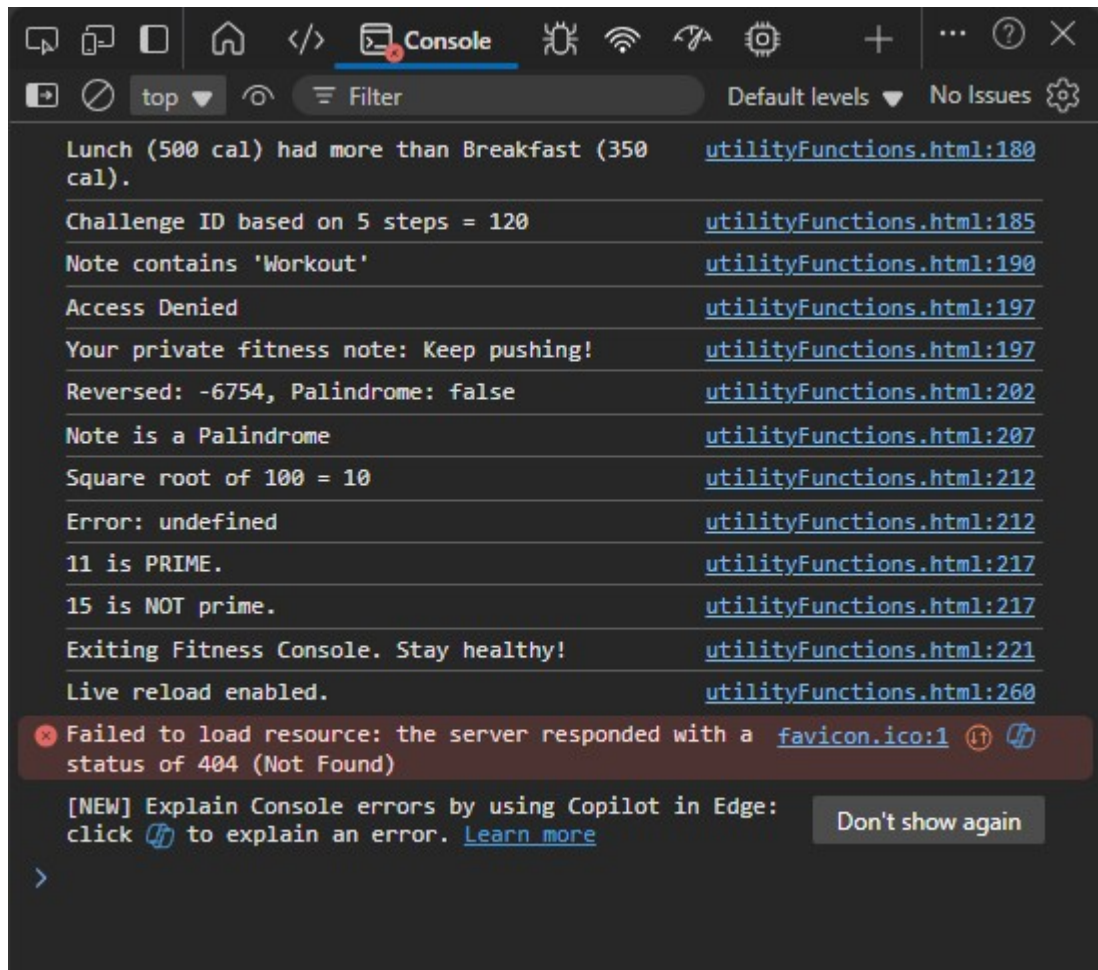
Code 10:



Code 11:



Code 12:



The screenshot shows a web browser's developer console with the 'Console' tab selected. The console displays a series of log messages from a JavaScript application, each followed by a link to the source file 'utilityFunctions.html' and a line number. The messages include calculations for calories, challenge IDs, note content, access status, fitness notes, palindrome checks, square roots, and prime number checks. At the bottom, there is a red error message indicating a failed resource load for 'favicon.ico' with a 404 status. A notification banner at the bottom of the console suggests using Copilot in Edge to explain errors, with a 'Don't show again' button.

```
Lunch (500 cal) had more than Breakfast (350 cal). utilityFunctions.html:180
Challenge ID based on 5 steps = 120 utilityFunctions.html:185
Note contains 'Workout' utilityFunctions.html:190
Access Denied utilityFunctions.html:197
Your private fitness note: Keep pushing! utilityFunctions.html:197
Reversed: -6754, Palindrome: false utilityFunctions.html:202
Note is a Palindrome utilityFunctions.html:207
Square root of 100 = 10 utilityFunctions.html:212
Error: undefined utilityFunctions.html:212
11 is PRIME. utilityFunctions.html:217
15 is NOT prime. utilityFunctions.html:217
Exiting Fitness Console. Stay healthy! utilityFunctions.html:221
Live reload enabled. utilityFunctions.html:260
Failed to load resource: the server responded with a status of 404 (Not Found) favicon.ico:1
[NEW] Explain Console errors by using Copilot in Edge: click to explain an error. Learn more
Don't show again
```